CUSTOMER SEGMENTATION ANALYSIS USING K-MEANS CLUSTERING

Data Analytics Internship Project Report

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ABSTRACT

Customer segmentation is an important technique used by businesses to understand customer behavior and improve marketing strategies. This project analyzes customer data using the K-Means Clustering algorithm to identify different groups of customers based on their annual income and spending score.

The objective of this project is to divide customers into meaningful segments so that businesses can target them with personalized marketing campaigns and improve customer satisfaction.

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1. INTRODUCTION

In today's competitive business environment, understanding customer behavior is essential for maximizing sales and customer retention.

Customer Segmentation is the process of dividing customers into groups based on similar characteristics such as:

- Age
- Income
- Spending habits
- Purchase behavior

Machine Learning algorithms such as K-Means Clustering help businesses automatically identify customer groups and generate valuable business insights.

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2. OBJECTIVE

The main objectives of this project are:

- Analyze customer data.
- Understand customer spending patterns.
- Apply K-Means Clustering.
- Group customers into meaningful segments.
- Generate business recommendations based on customer behavior.

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3. DATASET DESCRIPTION

The dataset used in this project is the Mall Customers Dataset.

Dataset Features

Feature	Description
CustomerID	Unique customer identifier
Gender	Male or Female
Age	Customer age
Annual Income (k$)	Annual income in thousand dollars
Spending Score (1-100)	Spending behavior score

Dataset Information

- Total Records: 200
- Total Columns: 5
- Missing Values: 0

(Insert Screenshot 1: Dataset Preview)

(Insert Screenshot 2: Dataset Information and Statistical Summary)

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4. TOOLS AND TECHNOLOGIES USED

Programming Language

- Python

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Development Environment

- Jupyter Notebook
- Anaconda

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5. METHODOLOGY

The project was completed in the following steps:

Step 1: Data Collection

The Mall Customers dataset was loaded into Python using Pandas.

Step 2: Data Exploration

Basic dataset information was analyzed using:

- df.head()
- df.shape
- df.info()
- df.describe()

Step 3: Data Visualization

Several visualizations were created to understand customer behavior:

- Gender Distribution
- Age Distribution
- Annual Income Distribution
- Spending Score Distribution
- Income vs Spending Scatter Plot

Step 4: Customer Segmentation

K-Means Clustering algorithm was applied on:

- Annual Income
- Spending Score

Step 5: Business Insights

Customer groups were analyzed and interpreted to provide marketing recommendations.

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6. PROJECT WORKFLOW

Dataset
↓
Data Exploration
↓
Data Visualization
↓
Feature Selection
↓
K-Means Clustering
↓
Customer Segmentation
↓
Business Insights

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7. IMPLEMENTATION

Importing Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
```

Loading Dataset

```
df = pd.read_csv("Mall_Customers.csv")
df.head()
```

Feature Selection

```
X = df[['Annual Income (k$)',
      'Spending Score (1-100)']]
```

Applying K-Means Clustering

```
kmeans = KMeans(
   n_clusters=5,
   random_state=42,
   n_init=10
)

y_kmeans = kmeans.fit_predict(X)
```

Creating Clusters

```
df['Cluster'] = y_kmeans
```

Visualizing Customer Segments

```
plt.scatter(
   X.iloc[:,0],
   X.iloc[:,1],
   c=y_kmeans
)
```

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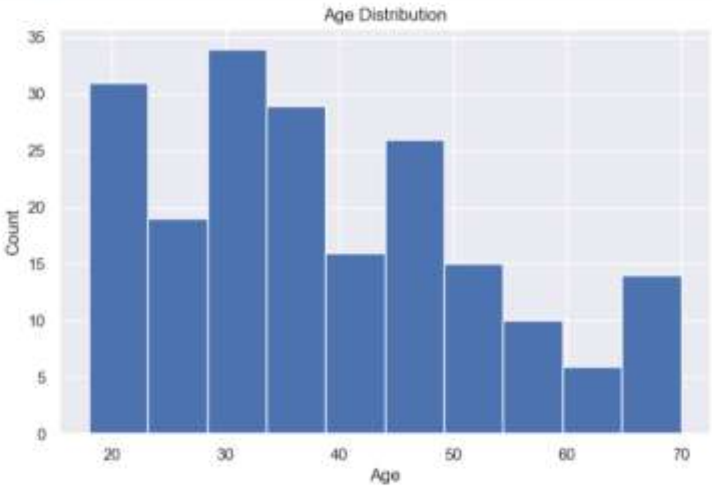
8. DATA VISUALIZATION RESULTS

Gender Distribution

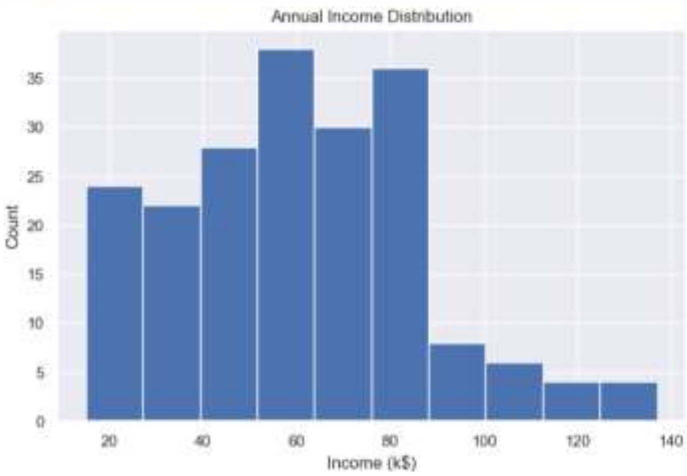
Age Distribution

```
[14]: plt.figure(figsize=(8,5))

      plt.hist(df['Age'], bins=10)

      plt.title('Age Distribution')
      plt.xlabel('Age')
      plt.ylabel('Count')

      plt.show()
```



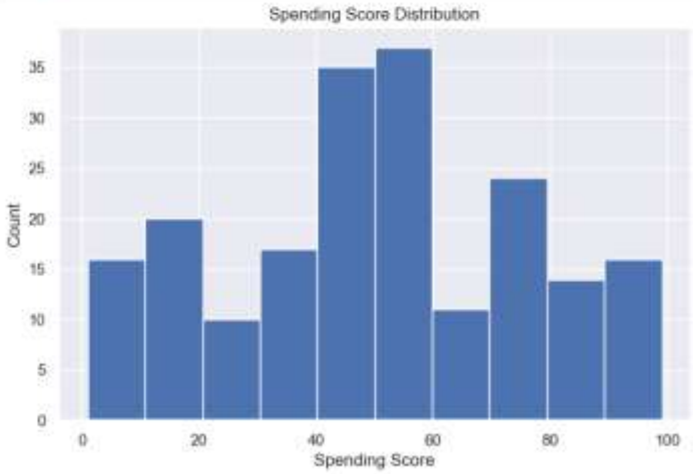
Annual Income Distribution

```
[15]: plt.figure(figsize=(8,5))

      plt.hist(df['Annual Income (k$)'], bins=10)

      plt.title('Annual Income Distribution')
      plt.xlabel('Income (k$)')
      plt.ylabel('Count')

      plt.show()
```



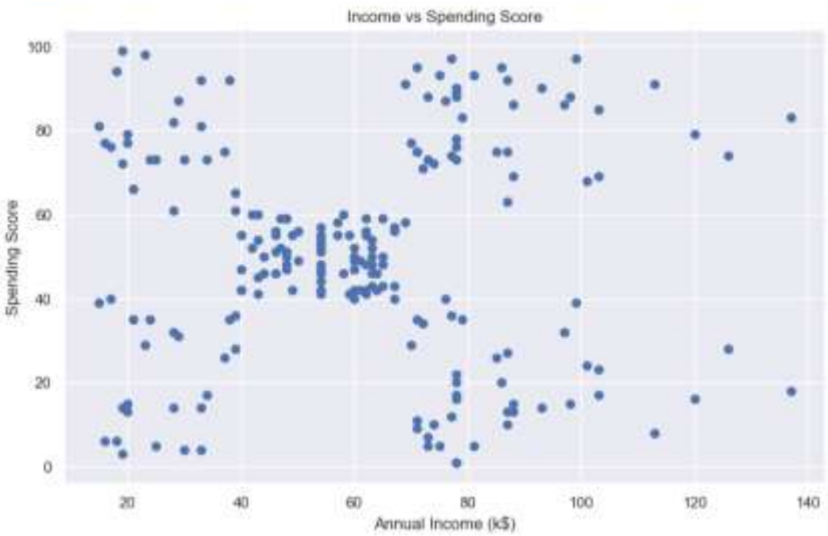
Spending Score Distribution

```
[16]: plt.figure(figsize=(8,5))

plt.hist(df['Spending Score (1-100)'], bins=10)

plt.title('Spending Score Distribution')
plt.xlabel('Spending Score')
plt.ylabel('Count')

plt.show()
```



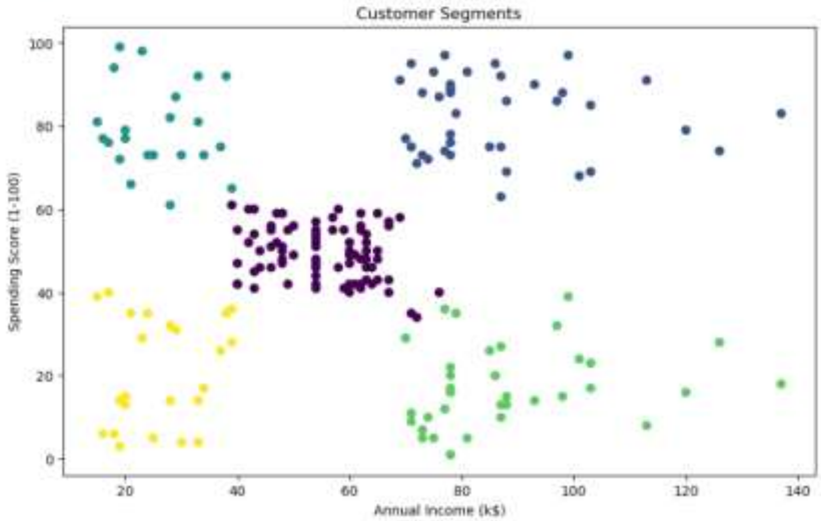
Income vs Spending Score

```
[17]: plt.figure(figsize=(10,6))

plt.scatter(
    df['Annual Income (k$)'],
    df['Spending Score (1-100)']
)

plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score')

plt.title('Income vs Spending Score')

plt.show()
```



Customer Segmentation Graph

```
[11]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

df = pd.read_csv(r"C:\Users\dadis\OneDrive\Documents\DataAnalytics\CustomerSegmentation\Mall_Customers.csv")

X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X)

plt.figure(figsize=(10,6))
plt.scatter(X.iloc[:,0], X.iloc[:,1], c=y_kmeans)

plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Customer Segments')

plt.show()
```



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9. CLUSTER ANALYSIS

Customer Distribution

Cluster	Customers
0	81
1	39
2	22
3	35
4	23

Average Values of Each Cluster

Cluster	**Annual Income**	**Spending Score**	**Age**
0	55.30	49.52	42.72
1	86.54	82.13	32.69
2	25.73	79.36	25.27
3	88.20	17.11	41.11
4	26.30	20.91	45.22

```
df.groupby('Cluster')[['Annual Income (k$)',
                       'Spending Score (1-100)',
                       'Age']].mean()
```

Cluster	Annual Income (k$)	Spending Score (1-100)	Age
0	55.296296	49.518519	42.716049
1	86.538462	82.128205	32.692308
2	25.727273	79.363636	25.272727
3	88.200000	17.114286	41.114286
4	26.304348	20.913043	45.217391

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10. BUSINESS INSIGHTS

Cluster 0

Average Income – Average Spending

- Regular customers
- Suitable for loyalty programs

Cluster 1

High Income – High Spending

- Premium customers
- Most valuable customer group
- Target with luxury products and exclusive offers

Cluster 2

Low Income – High Spending

- Young enthusiastic shoppers
- Respond well to discounts and offers

Cluster 3

High Income – Low Spending

- Potential customers
- Can be converted through personalized marketing

Cluster 4

Low Income – Low Spending

- Budget-conscious customers
- Suitable for affordable product campaigns

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11. CONCLUSION

Customer Segmentation Analysis was successfully performed using the K-Means Clustering algorithm.

Five distinct customer groups were identified based on Annual Income and Spending Score. The analysis revealed different customer behaviors and spending patterns, enabling businesses to design targeted marketing strategies.

This project demonstrates how Data Analytics and Machine Learning can be used to transform raw customer data into meaningful business insights.

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12. FUTURE ENHANCEMENTS

Future improvements can include:

- Using larger real-world datasets.
- Applying advanced clustering algorithms.
- Developing interactive dashboards using Power BI.
- Integrating customer purchase history.
- Real-time customer segmentation systems.

REFERENCES

1. Python Documentation
2. Scikit-Learn Documentation
3. Pandas Documentation
4. Matplotlib Documentation
5. Mall Customers Dataset (Kaggle)